STUDENT RECORD AND ACADEMIC MANAGEMENT SYSTEM

PYTHON SOURCE CODE

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Course: FY BTech – Artificial Intelligence and Data Science

Subject: Introduction to Programming and Data Structures

Complete menu-driven Python 3 program demonstrating strings, lists, tuples, sets, dictionaries, functions, loops, conditionals and input validation.

Run: Save as `student_record_system.py` and execute with Python 3.

Complete Python Code

```
"""Student Record and Academic Management System - Python 3"""
students = {}  # student ID -> (name, department, marks)

def add_student():
    sid = input("Student ID: ").strip()
    if not sid or sid in students:
        print("ID is empty or already exists."); return
    name = input("Student name: ").strip()
    dept = input("Department: ").strip()
    try:
        marks = float(input("Marks (0-100): "))
        if not 0 <= marks <= 100: raise ValueError
    except ValueError:
        print("Enter numeric marks from 0 to 100."); return
    students[sid] = (name, dept, marks)
    print("Student added.")

def search_student():
    sid = input("Student ID to search: ").strip()
    if sid in students:
        name, dept, marks = students[sid]
        print(f"ID: {sid} | Name: {name} | Department: {dept} | Marks: {marks:g}")
    else: print("Student not found.")

def update_student():
    sid = input("Student ID to update: ").strip()
    if sid not in students:
        print("Student not found."); return
    old_name, old_dept, old_marks = students[sid]
    name = input(f"Name [{old_name}]: ").strip() or old_name
    dept = input(f"Department [{old_dept}]: ").strip() or old_dept
    raw = input(f"Marks [{old_marks:g}]: ").strip()
    if raw:
        try:
            marks = float(raw)
            if not 0 <= marks <= 100: raise ValueError
        except ValueError:
            print("Invalid marks; update cancelled."); return
    else: marks = old_marks
    students[sid] = (name, dept, marks)
    print("Record updated.")

def delete_student():
    sid = input("Student ID to delete: ").strip()
    if sid in students:
        del students[sid]
        print("Record deleted.")
    else: print("Student not found.")

def display_all():
    if not students:
        print("No student records."); return
    for sid, (name, dept, marks) in students.items():
        print(f"ID: {sid} | Name: {name} | Department: {dept} | Marks: {marks:g}")

def average_marks():
    if not students:
        print("No records to calculate average."); return
    avg = sum(record[2] for record in students.values()) / len(students)
    print(f"Average marks: {avg:.2f}")
```

Complete Python Code (continued)

```
def highest_scorer():
    if not students:
        print("No student records."); return
    sid, record = max(students.items(), key=lambda item: item[1][2])
    print(f"Highest scorer: {record[0]} (ID {sid}), {record[2]:g} marks")

def list_by_department():
    dept = input("Department to filter: ").strip().casefold()
    matches = [(sid, r) for sid, r in students.items()
               if r[1].casefold() == dept]
    if not matches:
        print("No students found in that department."); return
    for sid, (name, department, marks) in matches:
        print(f"ID: {sid} | Name: {name} | Department: {department} | Marks: {marks:g}")

def show_departments():
    departments = {record[1] for record in students.values()}
    print("Departments:", ", ".join(sorted(departments)) if departments else "None")

def main():
    while True:
        print("\\n=== STUDENT RECORD AND ACADEMIC MANAGEMENT SYSTEM ===")
        print("1 Add  2 Search  3 Update  4 Delete  5 Display all")
        print("6 Average marks  7 Highest scorer  8 By department")
        print("9 Unique departments  0 Exit")
        choice = input("Choose option: ").strip()
        if choice == "1": add_student()
        elif choice == "2": search_student()
        elif choice == "3": update_student()
        elif choice == "4": delete_student()
        elif choice == "5": display_all()
        elif choice == "6": average_marks()
        elif choice == "7": highest_scorer()
        elif choice == "8": list_by_department()
        elif choice == "9": show_departments()
        elif choice == "0":
            print("Thank you."); break
        else: print("Invalid choice.")

if __name__ == "__main__":
    main()
```